Sarfraz Ahmad

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Sarfraz Ahmad | SarfrazAhmad307 | Portfolio | Location: UAE

Research Focus

NLP researcher with 5 years of experience in AI, machine learning, and Large Language Models (LLMs). Experienced in fact-checking, temporal and spatial reasoning, knowledge extraction, and model evaluation. Strong background in deep learning and transformers, with a research interest in explainable and trustworthy AI, particularly improving the interpretability and reliability of NLP systems.

Education

- **Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)** *Sep.2024 - May.2026*
  M.S. Natural Language Processing Abu Dhabi, United Arab Emirates
  - **GPA:** 3.83/4.00
  - **Supervisor:** Prof. Preslav Nakov
  - **Thesis:** Advancing Fact-Checking in Low-Resource and Multilingual Contexts: Benchmarks, Frameworks, and Evaluations
- **University of Engineering & Technology (UET)** *Oct.2016 - Jul.2020*
  B.S. Electrical Engineering (Computer Major) Lahore, Pakistan
  - **GPA:** 3.49/4.00
  - **Advisor:** Dr. Bilal Wajid Abbas
  - **Capstone Project:** Employing Machine Learning to understand Drug Mechanisms in Diseases

Work Experience

- **GoPython Software Solutions, LLC** *Jun. 2021 – July. 2024*
  Lead Data Scientist for Prognos Health Inc Chatham, NJ, United States of America
  - Collaborated with clinical experts to design rule-based transformation pipelines, enabling actionable health insights.
  - Deployed and optimized workflows on Databricks to enhance ETL performance and reliability.
  - Managed technical teams to ensure quality assurance, data governance, and efficient project delivery.
  - Developed AWS Lambda–triggered frameworks for large-scale automated web file updates and scraping tasks.
  - Built a Vue3 web app for querying medical databases, significantly improving data accessibility.
- **GoPython Software Solutions, LLC** *Nov. 2020 – May. 2021*
  Front-End Developer for Instant Labs Inc Chatham, NJ, United States of America
  - Developed an interactive drag-and-drop web app in React.js for designing complex database schemas.
  - Created a dynamic SQL generation tool using Blockly and React.js for real-time query execution.
  - Engineered backend RESTful APIs in Go to support seamless client–server communication.

Research Experience

- **Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)** *Jun. 2026 – Present*
  Research Engineer I Abu Dhabi, UAE
  - Conducting research on **fact-checking and multimodal fact-checking**, with a focus on evaluating and improving the factual reliability of Large Language Models (LLMs) and Vision-Language Models (VLMs).
- **Singapore University of Technology and Design (SUTD)** *Aug. 2025 – Nov. 2025*
  Masters Research Intern under Prof. Roy Ka-Wei Lee Singapore
  - Co-developed **TABVERSE**, a large-scale benchmark aligning over 10,000 tables across HTML, Markdown, and LaTeX formats, along with their rendered images, enabling direct comparison of Large Language Models (LLMs) and Vision-Language Models (VLMs).
- **Data Science Lab, Al-Khawarizmi Institute of Computer Science** *Jan. 2020 – Sep. 2020*
  Research Officer Lahore, Pakistan
  - Developed models for Explainable AI (XAI) to improve interpretability and transparency of deep learning systems.
  - Automated clinical coding workflows via machine learning, increasing accuracy and reducing manual effort.
  - Applied deep learning for injury classification, achieving high diagnostic accuracy across datasets.
  - Optimized algorithms such as PSO, PCA, and SVD to reduce model complexity and enhance performance.

- **Data Science Lab, Al-Khawarizmi Institute of Computer Science** *Oct. 2019 – Dec. 2019*
  Research Intern Lahore, Pakistan
  - Benchmarked activation functions and optimizers to identify optimal configurations for neural models.
  - Applied feature engineering techniques to improve data representation and model performance.
- **Sabz Qalam** *Dec. 2018 – Jun. 2020*
  Undergrad Research Intern Lahore, Pakistan
  - Contributed to oncology research in Metastatic Breast Cancer and Uveal Melanoma.
  - Applied advanced research methodologies to support data-driven clinical studies.

PUBLICATIONS

C=CONFERENCE, W=WORKSHOP, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T=THESIS

[C.1] **UrduMMMLU: A Massive Multitask Benchmark for Urdu Language Understanding**.
Tabassum, A.*, **Ahmad, S.***, Iqbal, H.*, Aijaz, O., Ahsan, M., & Nakov, P.
Proceedings of The 2026 Conference on Empirical Methods in Natural Language Processing (Long Papers) *Budapest, Hungary — 2026*
DOI: 2606.07167

[C.2] **MedSNIP: Building and Benchmarking Snippet-Level Granularity for Medical Fact Verification**.
Iqbal, H., **Ahmad, S.**, Kim, H., Park, S., Liao, J., Chen, Q., Nakov, P., & Wang, Y.
Proceedings of The 2026 Conference on Empirical Methods in Natural Language Processing (Long Papers) *Budapest, Hungary — 2026*
DOI: 2609.12884

[C.3] **Aligning Cultural Behaviors in Arabic Language Models via Preferences**.
Liang, J., Orel, D., Almheiri, S., Elbouardi, B., Samih, Y., Freihat, A. A., Xie, Z., Ahsan, M., Al Kautsar, M. D., **Ahmad, S.**, Keleg, A., El Herraoui, O., Anwar, M., Nakov, P., & Koto, F.
Proceedings of The 2026 Conference on Empirical Methods in Natural Language Processing (Long Papers) *Budapest, Hungary — 2026*
Available at: Google Drive Link

[C.4] **Jais 2: A Family of Arabic-Centric Open Large Language Models**.
Anwar, M., Freihat, A. A., Ibrahim, G., . . . , **Ahmad, S.**, . . . , & Nakov, P.
Proceedings of The 2026 Conference on Empirical Methods in Natural Language Processing (Long Papers) *Budapest, Hungary — 2026*
DOI: 2608.13580

[C.5] **SAHM: A Holistic Arabic Financial Benchmark Bridging Conventional Markets and Islamic Law**.
Elbadry, R., **Ahmad, S.**, Bouch, D., Ahsan, M., Peng, X., Huang, J., Anwar, M., AlMahri, M., Elsaid Khalil, M., Wang, Y., Lahlou, S., Stoyanov, V., Ananiadou, S., Nakov, P., & Xie, Z.
Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Long Papers) *San Diego, California, United States — 2026*
DOI: 10.18653/v1/2026.acl-long.1593

[C.6] **Cultural Benchmarking of LLMs in MSA and Arabic Dialectal Dialogue**.
Al Kautsar, M. D., Almheiri, S., Ahsan, M., Elbouardi, B., Samih, Y., **Ahmad, S.**, Keleg, A., El Herraoui, O., Elzeky, K., Freihat, A. A., Anwar, M., Xie, Z., Liang, J., Al Nasar, M. R., Nakov, P., & Koto, F.
Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Long Papers) *San Diego, California, United States — 2026*
DOI: 10.18653/v1/2026.acl-long.963

[C.7] **ImageCLEF 2026: Multimodal Challenges in Medicine, Science, Agritech, and Security**.
In European Conference on Information Retrieval, pp. 336–344. Cham *Springer Nature Switzerland — 2026*
DOI: 10.1007/978-3-032-21321-1_44

[C.8] **Overview of ImageCLEF 2026: Multimodal Challenges in Medicine, Science, Agritech, and Security**.
In Conference and Labs of the Evaluation Forum (CLEF 2026), pp. 465–492. Cham *Springer Nature Switzerland — 2027*
DOI: 10.1007/978-3-032-39150-6_29

[C.9] **A Parallel Cross-Lingual Benchmark for Multimodal Idiomaticity Understanding**.
Torunoğlu-Selamet, D., Arslan, D., Wilkens, R., . . . , **Ahmad, S.**, . . . , & Xie, Z.
LREC-COLING 2026 *Palma de Mallorca — 2026*
DOI: 10.63317/5cvnbcoktfo2

[C.10] **UrduFactCheck: An Agentic Fact-Checking Framework for Urdu with Evidence Boosting and Benchmarking**.
Ahmad, S.*, Iqbal, H.*, Ahsan, M., Naeem, N., Khan, M., Riaz, A., Manzoor, M., Wang, Y., & Nakov, P.
Findings of the Association for Computational Linguistics: EMNLP 2025 *Suzhou, China — 2025*
DOI: 10.18653/v1/2025.findings-emnlp.1240

[W.1] **Overview of the ImageCLEF 2026 Task on Multimodal Reasoning**.
Dimitrov, D., Hee, M. S., Ahsan, M., **Ahmad, S.**, Pachov, G., Zlatkova, D., Xie, Z., Nakov, P., & Koychev, I.,
CLEF 2025 Working Notes *Jena, Germany — Sep 21–24, 2026*
DOI: Vol-4283/paper235

[W.2] **Overview of ImageCLEF 2025 – Multimodal Reasoning**.
Dimitrov, D., Hee, M. S., Xie, Z., Das, R. J., Ahsan, M., **Ahmad, S.**, Paev, N., Koychev, I., & Nakov, P.
CLEF 2025 Working Notes *Madrid, Spain — Sep 9–12, 2025*
DOI: Vol-4038/paper_174

[W.3] **NeuralNexus at BEA 2025 Shared Task: Retrieval-Augmented Prompting for Mistake Identification in AI Tutors**.
Naeem, N., **Ahmad, S.**, Ahsan, M., & Iqbal, H.
Proceedings of the 20th Workshop on Innovative Use of NLP for Building Educational Applications (BEA 2025) *Vienna, Austria — 2025*
DOI: 10.18653/v1/2025.bea-1.100

[J.1] **iBitter-Stack: A Multi-Representation Ensemble Learning Model for Accurate Bitter Peptide Identification**.
Ahmad, S., Ahsan, M., Asim, M. N., Dengel, A., & Malik, M. I.
Journal of Molecular Biology, Vol. 437, No. 24, Article 169448 *— 2025*
DOI: 10.1016/j.jmb.2025.169448

[J.2] **Unearthing Insights into Metabolic Syndrome by Linking Drugs, Targets, and Gene Expressions Using Similarity Measures and Graph Theory**.
Zafar, A., Wajid, B., Shabbir, A., Awan, F. G., Ahsan, M., **Ahmad, S.**, Wajid, I., Anwar, F., & Mazhar, F.
Current Computer-Aided Drug Design, Vol. 20, Issue 6, pp. 773–783 *— 2024*
DOI: 10.2174/1573409920666230817101913

[B.1] **Linux Programming for the Faint of Heart**.
Sobahi, N., Al-Rashed, F., Shabbir, A., **Ahmad, S.**, Jamil, M., & Wajid, B.
Sabz Qalam Publications, Vol. 1, pp. 79–87. ISBN: 978-969-7941-00-1 *— 2020*

[S.1] **AraCuisine: Culturally Grounded Instruction Tuning and Benchmarking for Arabic Culinary Reasoning**.
Freihat, A. A., Samih, Y., Awad, M., Anwar, M., **Ahmad, S.**, Atif, F., Ahsan, M., Ibrahim, G., Koto, F., & Nakov, P.
ACL Rolling Review (ARR), Under Review *October 2026 Cycle — 2026*
Available at: Google Drive Link

[S.2] **ONEIROBENCH: A Source-Grounded Trilingual Benchmark for Dream Interpretation Across Textual Traditions**.
Iqbal, H.*, **Ahmad, S.***, M., Ahsan, Mekky, A., Wang, , M., Freihat, A. A., Anwar, M., Xie, Z., Nakov, P., & Wang, Y.
ACL Rolling Review (ARR), Under Review *October 2026 Cycle — 2026*
Available at: Google Drive Link

[S.3] **TABVERSE: Benchmarking Cross-Format Table Understanding in LLMs and VLMs**.
Ahsan, M., **Ahmad, S.**, Hee, M. S., Lee, R. K.-W., & Nakov, P.
ACL Rolling Review (ARR), Under Review *October 2026 Cycle — 2026*
DOI: 2606.09578

[S.4] **Open-VEQA: A Multilingual Multimodal OpenQA Benchmark for Educational Reasoning**.
Dimitrov, D. I., Ahsan, M., **Ahmad, S.**, Yotkova, E., Kastreva, V., Hee, M. S., Koychev, I., & Nakov, P.
ACL Rolling Review (ARR), Under Review *October 2026 Cycle — 2026*
Available at: Google Drive Link

[S.5] **PRIMEAPE: A Multilingual Benchmark for Fine-grained Meta-evaluation of Judge Models**.
Mikhailov, V., **Ahmad, S.**, Ahsan, M., Butenko, Z., Dobrovolskii, V., Nakov, P., Velldal, E., Øvrelid, L., & Artemova, E.
ACL Rolling Review (ARR), Under Review *October 2026 Cycle — 2026*
Available at: Google Drive Link

PROJECTS

- **Primeape: Multilingual Preference Prediction and Explanation Benchmark** *2025*
  Research Contributor MBZUAI
  - Developed a multilingual benchmark for preference prediction and natural language explanation generation across seven low-resource languages, including Urdu, Hindi, Yoruba, Norwegian, French, and Spanish, using human-annotated response comparisons.
- **Arabic Cuisine Benchmark (ACB): Cultural Reasoning in Arabic Food** *2025*
  Research Contributor MBZUAI
  - Developed the first large-scale benchmark for cultural reasoning in Arabic cuisine, comprising over 65K recipes across 20+ Arab countries and 1,300 validated multiple-choice questions.
- **Automatic Evaluation of Structured Text Using LLMs** *2025*
  Semester Project for course "Advanced Natural Lanuguage Processing-702" MBZUAI
  - Investigated the limitations of traditional metrics like ROUGE and entity-based measures in evaluating long-form, structured text generated by models.
  - Leveraged LLMs to parse and compare structural and logical elements of AI-generated articles against expert-written ground truths, revealing deeper insights into content quality.

◦ Proposed evaluation strategies for structure-aware assessment at node, argument, and claim levels, enabling fine-grained analysis of logical flow and factual consistency.

- **AI-Driven Framework for Identifying COVID-19 Severity Using Single-Cell Data** *2025*
  Semester Project for course "Introduction to Single Cell Biology and Bioinformatics-703" MBZUAI
  ◦ Developed a machine learning pipeline leveraging single-cell RNA sequencing (scRNA-seq) data to classify COVID-19 severity at both cell and patient levels.
- **Leveraging LLMs for Digital Discourse Detection and Analysis in Multilingual Contexts** *2024*
  Semester Project for course "Foundations of Artificial Intelligence-701" MBZUAI
  ◦ This study evaluates hate speech detection in German, Hindi, and English using both traditional ML models and advanced LLMs, demonstrating the superiority of LLMs in low-resource settings.
- **Leveraging Natural Language Processing for Peptide Sequence Classification and Insight Discovery** *2024*
  Semester Project for course "Natural Lanuguage Processing-701" MBZUAI
  ◦ The project utilized LLM embeddings and CNNs in an ensemble framework to classify peptides, capturing sequence-specific patterns without handcrafted features.
- **Incorporating interpretability in ML/DL models using Shapley values and SHAP** *2022*
  Freelance Project Freelancing
  ◦ Studied and incorporated the concepts of corporate game theory and shapley values.
  ◦ Utilized the aforementioned concepts to induce interpretability in sentiment analysis task using deep learning.
- **Breast Cancer Histopathological Image Classification using Transfer Learning** *2022*
  Freelance Project Freelancing
  ◦ Utilized transfer learning-based model, DenseNet-201, and fin-tuned the pre-trained model for the task of binary classification of Breast Histopathology images.
  ◦ The experimentation was performed on the basis of magnification factor. For images at each magnification level, we trained 5 models on 5 folds provided by the creators of this dataset and accuracy for these model were averaged for final results.
- **Employing Machine Learning to understand Drug Mechanisms in Diseases** *2020*
  Capstone Project of Under-graduation UET
  ◦ Classified Breast Cancer Metastases using optimal classifiers, with preprocessing, filtering, and filter-based feature selection to identify key genes.

Skills

- **Programming Languages:** Python, C++, Java, JavaScript, Scala, Go
- **Web Technologies:** HTML, CSS, Angular, React, Node.js, Vue3
- **Database Systems:** MySQL, PostgreSQL, MongoDB
- **Data Science & Machine Learning:** TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Keras, Selenium, OpenCV, Scrappy
- **Cloud Technologies:** AWS (Athena, ECS, S3, Lambda), Microsoft Azure, Databricks
- **DevOps & Version Control:** Docker, Terraform, Kubernetes, Git, GitHub
- **Specialized Area:** Natural Language Processing, Deep Learning, Time Series Forecasting, Biomedical Data Analysis
- **Mathematical & Statistical Tools:** MATLAB, R, SAS, SPSS, Excel
- **Other Tools & Technologies:** Apache Spark, Tableau, JIRA
- **Research Skills:** Literature Review, Experimental Design, Data Analysis, Statistical Modeling, Hypothesis Testing, Academic Writing

Certifications

- **Cousera** Completed 65 hours course named "Machine Learning" offered by Stanford Online *2020*
- **Cousera** Completed 24 hours course "Neural Networks & Deep Learning" offered by Deeplearning.ai *2020*
- **Cousera** Completed "Improving Deep Neural Networks" course offered by Deeplearning.ai *2020*
- **Cousera** Completed 35-hour "Introduction to Programming with MATLAB" course by Vanderbilt University. *2020*
- **365 Data Science** Completed courses "Intro to Git & Github" *2020*
- **365 Data Science** Completed course "Introduction to Tableau" *2020*
- **365 Data Science** Completed course "Introduction to SQL" *2020*
- **World Health Organization** Participated in "Infection Prevention and Control (IPC) for Novel Coronavirus" *2020*
- **Microsoft** Got certification from Microsoft for completing MS Office Powerpoint Specialist test *2017*

References

Available upon request